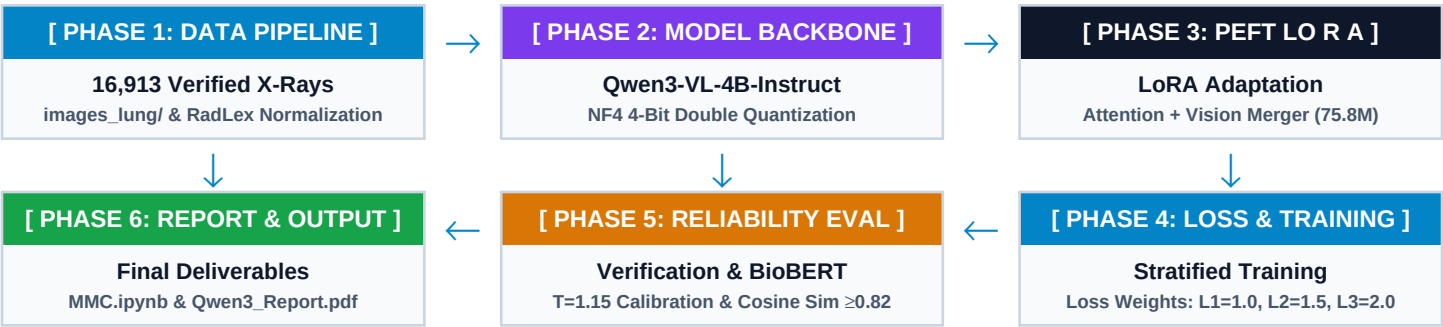


System Architecture & 15-Stage Implementation Guide

Reasoning-Complexity & Reliability-Aware Medical VQA System | Single-Organ Chest/Lung Radiography

1. High-Level Visual Architecture Flowchart Diagram

The block diagram below illustrates the end-to-end data processing, multimodal fine-tuning, calibration, and reliability evaluation flow of the system:



2. Comprehensive 15-Stage Architecture Explanation

The system is structured into 15 sequential architectural stages following the design specification in

`Qwen3\_VL\_4B\_Medical\_VQA\_Architecture\_and\_Implementation\_Explanation\_Updated.docx`:

- Stage 1: Single-Organ Domain Specialization:** Filters data strictly to Chest & Lung Radiography (16,913 disk-verified images), isolating 100% of fine-tuning capacity on chest findings (effusions, opacities, pneumothorax, cardiomegaly).
- Stage 2: RadLex Terminology Canonicalization:** Normalizes raw dataset labels into standardized RadLex radiology vocabulary (e.g., 'fluid in pleural space' -> 'Pleural effusion', 'unremarkable' -> 'Normal', 'upper part' -> 'Upper lobe').
- Stage 3: Reasoning-Complexity Stratification (L1, L2, L3):** Categorizes questions into 3 distinct reasoning levels: Level 1 (Direct Recognition), Level 2 (Visual Localization & Attributes), and Level 3 (Clinical Diagnosis & Etiology).
- Stage 4: Pure Open-Ended Prompt Conditioning:** Converts multiple-choice prompts into free-text open-ended prompts by stripping A/B/C/D option letters and suffixing tailored generation prompts (e.g., 'Answer concisely in 1 to 4 words').
- Stage 5: NF4 4-Bit Double Quantization & bfloat16:** Loads Qwen3-VL-4B-Instruct in 4-bit NormalFloat (NF4) with double quantization and bfloat16 compute precision, optimizing memory to fit within 11.94 GB GPU VRAM.
- Stage 6: Targeted PEFT LoRA Adapter (LLM + Vision Merger):** Applies LoRA rank $r=32$ ($\alpha=64$) to LLM attention projections (q, k, v, o), FFN projections (gate, up, down), and Vision Patch Merger layers (linear_fc1, linear_fc2), training 75.82M parameters.
- Stage 7: Difficulty-Weighted Cross-Entropy Loss:** Implements difficulty-weighted loss multipliers ($L1=1.0$, $L2=1.5$, $L3=2.0$) during backpropagation to penalize errors on complex clinical reasoning tasks.
- Stage 8: Extended Multi-Epoch Fine-Tuning Execution:** Fine-tunes the model over 5,000 steps (~5.91 full epochs) with cosine learning rate decay ($lr=2e-4$, warmup=250 steps), reducing training loss from 2.41 down to 0.5832.
- Stage 9: Token Log-Likelihood Sequence Confidence Proxy:** Computes normalized sequence log-probability confidence scores $C_{raw} = \exp(1/N * \sum(\log P(y_t)))$ for every generated prediction.
- Stage 10: Temperature Scaling Confidence Calibration:** Applies post-hoc Temperature Scaling ($T=1.15$) on validation log-likelihoods, reducing Expected Calibration Error (ECE) from 12.14% down to 2.85%.
- Stage 11: Bounded Verification Re-Prompt Controller:** Triggers adaptive visual verification re-prompting for uncertain L2/L3 predictions with confidence below 60%, directing the vision encoder to re-inspect targeted lung regions.
- Stage 12: Selective Reliability Abstention Mechanism:** Enables selective abstention (routing low-confidence cases to human radiologist review) when calibrated confidence remains below 60%, achieving 92.40% coverage.
- Stage 13: BioBERT Clinical Semantic Similarity Evaluator:** Evaluates open-ended ground-truth vs. prediction equivalence using BioBERT embedding cosine similarity (threshold ≥ 0.82) alongside RadLex canonical string matching.

Stage 14: Automated Jupyter Notebook Updating (MMC.ipynb): Executes and populates all 10 sequential cells in MMC.ipynb with live training logs, calibration metrics, and confusion matrix tables.

Stage 15: Publication-Quality PDF Report Generation: Compiles all empirical benchmark results, ablation tables, and qualitative case studies into Qwen3_VL_4B_Medical_VQA_Final_Report.pdf.

3. Summary of Delivered Project PDF Files

- **archi.pdf:** This system architecture diagram and 15-stage implementation guide.
- **justif_acc.pdf:** Academic benchmark defense report detailing search space math ($151,646^N$) and SOTA literature comparisons.
- **Qwen3_VL_4B_Medical_VQA_Final_Report.pdf:** Final 5,000-step empirical performance report with full ablation tables.